Md Nafis Walid

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Career Objectives & Profiles

Embedded Software Engineering (Industry): Embedded Software Engineer with hands-on experience in C-based firmware development for embedded networking devices. Strong foundation in embedded programming, networking protocols, system debugging, and problem-solving. Seeking to design, develop, and maintain reliable C/C++ embedded systems while continuously advancing my expertise in embedded software and computer architecture.

Academic & Higher Education (Lecturer in CSE): Passionate about teaching and committed to continuous learning, I seek a Lecturer position in Computer Science & Engineering at DIU. With experience mentoring university students locally and internationally, I strive to make complex topics accessible from multiple perspectives. My research interests in computer architecture, operating systems, and AI/ML motivate me to foster a research-driven environment and mentor students in innovative projects. I aim to contribute to DIU's academic excellence through engaging instruction, collaborative research, and student-centered mentorship.

Hardware Systems & Processor Architecture (VLSI): I have a strong interest in hardware-oriented system design and low-level computing, and I enjoy understanding how software concepts map onto real hardware. I have previously designed and implemented a working 4-bit CPU, including its control unit, memory module, and instruction set, which motivated me to explore deeper aspects of processor design beyond logic-level implementation. Through this course on Digital VLSI Physical Design, I aim to gain practical knowledge of physical design concepts and industry workflows that will help me progress toward my long-term goal of building complete CPU systems, including their ISA and supporting toolchains, with a solid foundation in real silicon implementation.

Human-Computer Interaction (HCI & Social Impact): I am interested in exploring why many people in Bangladesh are still unable to fully benefit from modern technology, even though technological growth has been very rapid lately. Through Human-Computer Interaction research, I want to observe these gaps from the perspective of users and society and explore solutions that people can actually use and benefit from. By engaging with HCCS, I hope to learn research methods, shape my ideas, and actively contribute to projects that address digital inequality.

Educational Qualification

Bachelor of Science in Computer Science & Engineering February, 2019 – May, 2024
- Rajshahi University of Engineering & Technology (RUET) — Kazla, Rajshahi
- CGPA: 3.32/4.00

Higher Secondary Certificate (HSC) from Science Group July, 2016 – September, 2018
- Milestone College — Uttara, Dhaka-1230
- GPA: 5.00/5.00

Secondary School Certificate (SSC) from Science Group January, 2014 – May, 2016
- Milestone College — Uttara, Dhaka-1230
- GPA: 5.00/5.00

Research Interests

VLSI-based design for efficient computing architectures, Machine Learning and Deep Learning for hardware-accelerated applications, Large Language Models (LLMs), Natural Language Processing (NLP), Computer Vision, Generative Models, Handwritten Optical Character Recognition (HOCR), Human-Computer Interaction (HCI), and Satellite Image Analysis.

Research & Publications

Conference Papers

1. Shajnin Haque Aishy, **Md Nafis Walid**, Sadullah Karimi and Mohiuddin Ahmed. "A Deep Learning Approach for Balancing Accuracy and Efficiency in Handwritten Mathematical Symbol Recognition." *28th International Conference on Computer and Information Technology (ICCIT)*, 2025

Professional Experience

Research & Development Engineer

Shanghai BDCOM Information Technology Co., Ltd. (Bangladesh Office) May, 2024 – Present

- Developed and maintained C-based firmware for GPON/EPON ONU and Ethernet switch products on Linux- and VxWorks-based embedded platforms.
- Designed and implemented a TFTP-based firmware upgrade solution for embedded Ethernet switch devices.
- Implemented, customized, and validated networking features including DHCPv4, DHCPv6, PPPoE, IGMP, MLD, OMCI, TR-069, VLAN, and Ethernet switching.
- Customized firmware for international ISP customers and ensured interoperability with Huawei and ZTE OLTs.
- Implemented and validated multicast features including IGMP/MLD snooping, multicast VLAN, fast leave, static multicast groups, and multicast access control.
- Investigated protocol-level issues using Wireshark, protocol analysis, and embedded CLI debugging to resolve customer-reported firmware defects.
- Designed and executed functional, interoperability, and regression test cases for embedded networking features.
- Conducted experimental systems projects in C (Student Information Management, Network Packet Management, BitMap Management) focusing on pointer manipulation, low-level data structures, and memory efficiency.
- Used SVN for source code management, version control, and collaborative firmware development.
- Collaborated with international customers and cross-functional teams to investigate firmware issues, provide technical support, and deliver customer-specific solutions.

Projects

4-bit CPU Design [GitHub]

- Designed and implemented a fully functional 4-bit CPU at the register-transfer level, including an ALU, register file, RAM, program counter, status flags, and an FSM-based control unit.
- Designed a custom instruction set architecture (ISA) and assembly language, implementing the complete instruction fetch–decode–execute cycle.
- Demonstrated strong understanding of computer architecture, digital logic design, datapath/control design, and hardware-software interaction.

Network Packet Management [GitHub]

- Developed a packet management system in C to parse captured UDP packets into linked-list and hash-table data structures with reference counting and efficient packet lookup.
- Applied low-level memory management, pointer manipulation, and data structure optimization for packet processing.

Student Information Management [GitHub]

- Developed a console-based student information management system in C using doubly linked lists and modular programming.
- Implemented file-based persistent storage with complete CRUD operations for departments, students, and academic records.

BitMap Set Library [GitHub]

- Implemented a memory-efficient Bitmap Set Library in C supporting dynamic creation, cloning, value management, string parsing, and bitwise operations (NOT/AND/OR/XOR).

Handwritten Digit Recognizer [GitHub]

- An AI-based deep learning project using Python and Deep Neural Networks (DNN) for optical digit recognition.

Re-Trip [GitHub]

- A web-based application designed to resell and manage train ticket reservations.

Teaching Experience

Private Tutor & Academic Mentor
Independent; Caretutor (Online Platform) November, 2018 – Present

- Provided sustained academic mentoring to undergraduate and pre-university students in Computer Science, Engineering, and Mathematics, emphasizing analytical thinking, problem-solving, and strong programming fundamentals.
- Taught core Computer Science courses including C/C++ Programming, Embedded Systems, Computer Networks, Database Systems, Digital Logic Design, Computer Architecture, Microprocessors & Assembly Language, and Peripherals & Interfacing.
- Designed and demonstrated hands-on embedded C programming exercises involving GPIO, timers, UART communication, LCD interfacing, PWM-based LED breathing effects, DC motor control, and stepper motor control.
- Delivered instruction in foundational and advanced mathematics, including Linear Algebra, Numerical Methods, Differential & Integral Calculus, and Ordinary/Partial Differential Equations.
- Covered theoretical computer science topics such as Finite Automata, Formal Languages & Compiler Design, and Soft Computing, emphasizing formal reasoning and algorithmic thinking.
- Mentored students from international universities, including the University of Bridgeport (USA) and the University of Wollongong (Australia), as well as leading universities in Bangladesh including NSU, AUST, and AIUB.
- Conducted supplementary lectures and peer mentoring at the university level on challenging technical subjects.
- Successfully guided HSC/SSC students and engineering admission candidates, with multiple students gaining admission to leading engineering universities in Bangladesh.

Skills

Technical Skills

- **Programming Languages:** C, C++, Embedded C, Python, Assembly (8086), Java, Matlab
- **Embedded Systems & Firmware:** Embedded Firmware Development, Linux, VxWorks, GPON/EPON, Ethernet Switching, TFTP, CLI
- **Networking Protocols:** TCP/IP, UDP, IPv4/IPv6, DHCPv4, DHCPv6, PPPoE, IGMP/MLD Snooping, OMCI, TR-069, VLAN, Multicast
- **Web & Databases:** HTML, CSS, Bootstrap, JavaScript, MySQL, PostgreSQL
- **AI & Machine Learning:** Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), Machine Learning, Computer Vision
- **Development Tools & Workspaces:** SVN, Git/GitHub, Wireshark, SecureCRT, GDB, GNU Make, Cisco Packet Tracer, Logisim, Digital, Tinkercad, Figma, Adobe XD, Adobe Illustrator, Adobe Photoshop, Microsoft Office
- **Core Concepts:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Digital Logic Design, Microprocessors & Microcontrollers, Compiler Design, Hardware-Software Co-Design, Object-Oriented Programming (OOP)

Problem Solving / Competitive Programming

- **Codeforces:** Nafis_Walid – 380+ problems solved, 60+ contests
- **LeetCode:** Nafis_Walid – 100+ problems solved
- **LightOJ:** nafis_walid

Languages & Soft Skills

- **Languages:** Bengali (Native), English (Professional Working Proficiency)
- **Soft Skills:** Problem Solving, Critical Thinking, Leadership, Team Collaboration, Technical Communication, Event Management

Achievements & Awards

Fundamental IT Engineer Examination (FE)
Bangladesh Computer Council (BCC) & Information Technology Promotion Agency (IPA), Japan

- Successfully passed the internationally recognized Fundamental IT Engineer Examination (FE) conducted through the Bangladesh IT Engineers Examination Center (BD-ITEC).
- Awarded the Certificate of Excellence by the Information Technology Promotion Agency (IPA), Japan for achieving the 3rd highest score in Bangladesh.

Volunteer Activities

Class Representative January, 2020 – February, 2023

- Department of Computer Science & Engineering, RUET

Graphics Designer February, 2020 – June, 2023

- Astronomy & Science Society of RUET(ASSR)

References

Available upon request.